Data Science

Lecture 03: Probabilistic Graphical Models

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Speaker notes

Converted from LNKB-tex part-en-ds-chpM1-graphical-models.tex. Prerequisite: Lecture 02 (probability theory) — in particular conditional independence, the chain rule and marginalisation, all of which this lecture uses continuously.

Where We Left Off

Lecture 02 gave us conditioning and Bayes for a handful of events. Real questions involve dozens of variables at once — and the assumption that makes them affordable says *which variables do not depend on each other*.

Lecture 02
Bayes for two events

→

Today
one joint distribution over many variables

→

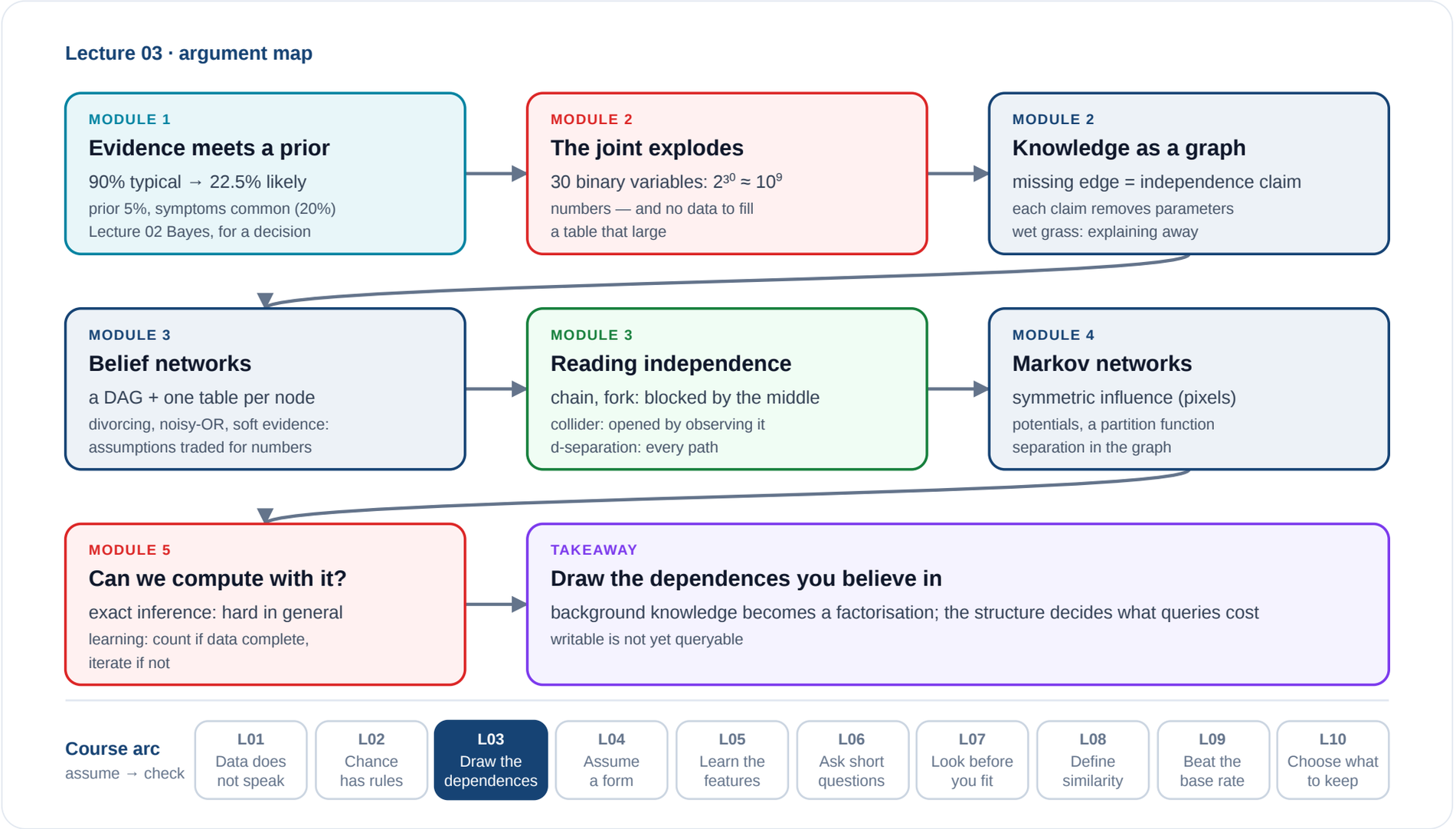
The assumption
a graph: every missing edge an independence claim

> Lecture 01's churn puzzle — does the app keep customers, or do loyal customers install the app? — is a question about arrows. Module 3 draws both answers.

Speaker notes

Module 1 (flu, 22.5%) is the second pass through Bayes. Keep it short and use it to show that the evidence term P(D) enters — the quantity a network later computes for us. The churn graph slide closes Module 3.

Argument Map



Speaker notes

The flu example shows Bayes serving a decision. The joint then explodes: 30 binary variables already need about 10^9 numbers. The graph is the way out: background knowledge, drawn as missing edges, removes parameters. Belief networks make it concrete; d-separation reads independence off the picture; Markov networks drop the direction. The last row is the catch: writing the joint down is not the same as being able to query it. Point at the strip: the assumption this lecture adds is the graph itself.

Module 1

Deciding Under Uncertainty

> ***Guiding Question:*** *If a symptom is 90% typical of a disease, why does having the symptom not make the disease 90% likely?*

Thinking With Uncertainty

Bayes' theorem lets us update a **hypothesis** H in the light of **data** D:

$$P(H \mid D) = \frac{P(H) \cdot P(D \mid H)}{P(D)}$$

This is the whole apparatus of evidence-based decision making in one line: a belief, a measurement of how well the belief explains what we saw, and a normalisation.

Speaker notes

Deliberate second pass over Bayes (P4/P5 decision rule: this is consolidation, so math first). The new element is the reading of H and D as hypothesis and data, which is the form in which learning algorithms use Bayes.

Example: Do I Have the Flu?

You have a headache and a cough. From the literature:

Likelihood

90% of people with the flu show headache and coughing

$$P(D \mid H) = 0.9$$

Prior

5% of people currently have the flu

$$P(H) = 0.05$$

Evidence

headache and coughing occur in 20% of people

$$P(D) = 0.2$$

How likely is it that you have the flu?

Speaker notes

Ask for a guess first — most people say something close to 90%, repeating the base-rate error from the probability chapter. 22.5% is the answer.

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Evidence

headache and coughing occur in 20% of people

$$P(D) = 0.2$$

How likely is it that you have the flu?

$$P(H \mid D) = \frac{0.05 \cdot 0.9}{0.2} = 0.225$$

Why So Low?

Few people have the flu

The prior $P(H) = 0.05$ caps how far the evidence can carry us.

The symptoms are common

$P(D) = 0.2$ means many people have these symptoms *without* the flu.

A symptom is informative to the degree that it is **rare in general** but **common under the hypothesis**. A likelihood alone tells you nothing.

Speaker notes

Two reasons, both structural. This is the criterion that reappears as feature informativeness in classification: a feature that fires for everything carries no information. Connect it explicitly — it is the idea behind information-based feature scores such as information gain.

Takeaway

Bayesian decision making weighs **prior × likelihood** against how ordinary the evidence is. Deciding under uncertainty means keeping all three numbers in view, not just the one the study reports.

Module 2

Why Graphical Models?

> ***Guiding Question:*** *Thirty binary variables need a billion numbers. What knowledge would let us write the same distribution with a few dozen?*

The Problem: Joint Distributions Explode

Consider a joint distribution $p(x_1, \ldots, x_n)$ over n **binary** random variables. How many configurations are there?



Speaker notes

Guiding question of the module. 30 binary variables already exceed a billion parameters. Ask what would have to be true of the world for the table to be compressible — the answer is independence, which the students already have.

The Problem: Joint Distributions Explode

Consider a joint distribution $p(x_1, \ldots, x_n)$ over n **binary** random variables. How many configurations are there?

$$2^n \quad \text{states}, \qquad 2^n - 1 \quad \text{free parameters}$$

Specification

For large n, writing down all values is impossible.

Inference

A single marginal $p(x_1)$ requires summing over 2^{n-1} states.

Chain rule?

It rewrites the joint exactly — but does not reduce the number of values.

What a Graphical Model Is

Definition (Graphical model). A graphical model depicts the dependence and independence relations of a distribution using a graph G as its underlying structure.

Modelling

Describe the variables of a domain and their interactions — how $p(x_1, \ldots, x_n)$ **factorises**.

Inference

Answer queries such as $p(x_1 \mid x_2 = a, \ldots)$ efficiently over that factorisation.

Usually one class of model is used for modelling and another for inference; the latter is often compiled from the former.

Speaker notes

Two purposes, two different graph classes — worth stating now so that Markov networks later do not look like a competing formalism but a compilation target.

How a Graph Can Help

Probabilistic inference is summing and multiplying. Let every variable be a node, and read edges as interaction:

No edge

$$f(a, b) = g(a) \cdot h(b)$$

The variables are independent — the factor splits.

Edge

$$f(a, b) \neq g(a) \cdot h(b)$$

The variables are dependent — the factor does not split.

Edge **direction** can encode further structure. The principle extends to any number of variables.

> Graphical models have limited expressive power: not every conditional independence statement can be drawn, and different graphs can represent the same distribution.

Speaker notes

The honest caveat matters — students should not believe the graph captures everything. "The graph is a picture of a factorisation" is the accurate slogan.

Example: Why Is the Grass Wet?

Tracey leaves her house and sees that her grass is wet. **Was it rain, or did she leave the sprinkler on?** She notices her neighbour Jack's grass is also wet.

R	S	J	T
raining (1) or not (0)	Tracey's sprinkler on (1) or off (0)	Jack's grass wet (1) or dry (0)	Tracey's grass wet (1) or dry (0)

Speaker notes

Barber's classic example. Ask students to predict, before any arithmetic, whether Jack's wet grass makes the sprinkler more or less likely — this sets up explaining away, and intuitions differ, which is the point.

Building the Model

The full joint over four binary variables needs $2^4 - 1 = 15$ probabilities. The chain rule rewrites it but does not shrink it:

$$p(T, J, R, S) = p(T \mid J, R, S)p(J \mid R, S)p(R \mid S)p(S)$$

Now bring in **background knowledge** — assumptions of conditional independence:

Tracey's grass does not depend on Jack's

$$p(T \mid J, R, S) = p(T \mid R, S)$$

Jack's grass does not depend on Tracey's sprinkler

$$p(J \mid R, S) = p(J \mid R)$$

Rain is not influenced by the sprinkler

$$p(R \mid S) = p(R)$$

$$p(T, J, R, S) = p(T \mid R, S)p(J \mid R)p(R)p(S)$$

Only $4 + 2 + 1 + 1 = 8$ probabilities remain — and each one is a question a person can actually answer.

Speaker notes

The heart of the module: the reduction is not mathematical trickery, it is domain knowledge written as conditional independence. The second half of the callout matters as much as the first — the factors are locally interpretable.

The Probability Tables

Factor	$p(\cdot)$	$1 - p(\cdot)$
$p(R = 1)$	0.2	0.8
$p(S = 1)$	0.1	0.9
$p(J = 1 \mid R = 0)$	0.2	0.8
$p(J = 1 \mid R = 1)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 0)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 1)$	1.0	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9	0.1
$p(T = 1 \mid R = 0, S = 0)$	0.0	1.0

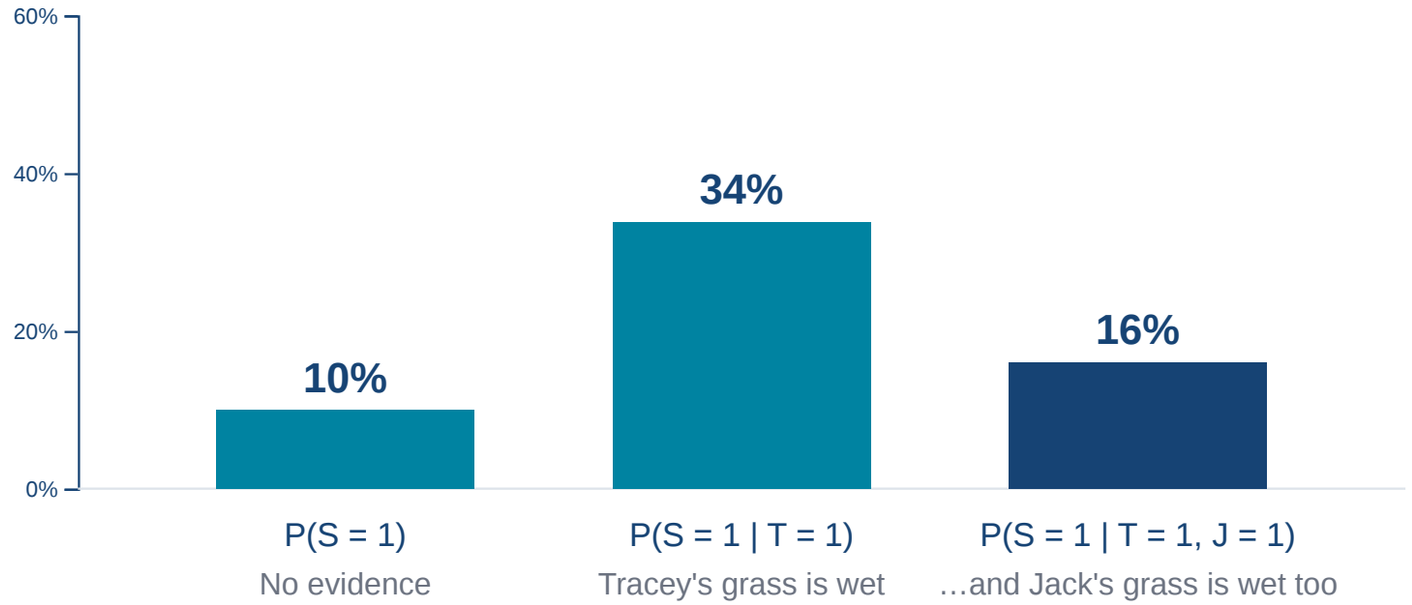
Missing edges in the graph are exactly the assumptions above; the tables supply the numbers.

Speaker notes

Point out that the last row encodes "grass does not get wet by itself" and the 1.0 entries encode "rain always wets grass" — modelling choices, visible and challengeable, which is the epistemic virtue of the approach (P3).

Inference: What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Speaker notes

Stepped viz (P6). Step 1: wet grass raises P(sprinkler) from 10% to 33.8%. Step 2: Jack's grass is wet too — rain now explains the wetness, and the sprinkler falls to 16.0%: most of the way back, but still above the 10% prior, because the sprinkler remains a possible additional cause. Ask for the prediction before each click. Computed by exact enumeration of the factorised joint, i.e. marginalisation the students already know.

Explaining Away

$$p(S=1 \mid T=1, J=1) = \frac{\sum_R p(T{=}1 \mid R, S{=}1)p(J{=}1 \mid R)p(R)p(S{=}1)}{\sum_{R,S} p(T{=}1 \mid R, S)p(J{=}1 \mid R)p(R)p(S)} = \frac{0.0344}{0.2144} \approx 0.16$$

Two independent causes (R and S) become **dependent once their common effect is observed**. Evidence for rain — Jack's wet grass — lowers the probability of the sprinkler from 0.34 to 0.16, even though rain and sprinkler are unconditionally independent.

Speaker notes

The formula is the second click of the viz, written out: the numerator keeps S = 1 (0.02 with rain, 0.0144 without), the denominator adds S = 0 (0.18 with rain, 0 without). This is the conditional-independence unit returning with a vengeance: conditioning on a common effect *creates* dependence — the opposite of the common-cause case seen earlier. Name it "explaining away"; it is the single most useful piece of vocabulary in this module.

Takeaway

A graphical model turns **background knowledge into a factorisation**: missing edges are conditional independence assumptions, and each one removes parameters. The graph is a picture of how the joint splits — nothing more, and nothing less.

Module 3

Bayesian Belief Networks

> ***Guiding Question:*** *What exactly does an arrow in a probabilistic model claim — and what does a missing arrow claim?*

Definition

Bayesian belief network (BBN). A **directed acyclic graph** whose nodes are random variables, in which the joint distribution factorises as

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p\big(x_i \mid \mathrm{pa}(x_i)\big)$$

where $\mathrm{pa}(x_i)$ are the parents of x_i in the graph.

Each node carries one conditional probability table given its parents. The wet-grass model was exactly this.

What the graph claims: each variable is independent of its non-descendants given its parents. A *missing* arrow is an independence claim; an arrow only permits a dependence.

Speaker notes

Math first (P5) — the students have already built one of these by hand, so the general form should read as a summary rather than a novelty. Stress "acyclic": cycles would make the factorisation ill-defined. The box is the local Markov property: it is what the factorisation means, read off the picture. In wet grass, Jack's lawn has only rain as a parent, so once rain is known the sprinkler tells us nothing more about it.

Three Tasks

Modelling

Choose variables, edges and tables that encode the domain.

Inference

Compute posteriors given evidence — at a cost set by the graph's shape.

Learning

Estimate the tables — and possibly the structure — from data.

Speaker notes

Say only that tractability depends on the *shape* of the network, not on its size alone: trees are easy, dense graphs are not. How hard inference is in general comes with inference itself.

Modelling Technique: Divorcing Parents

A node with k binary parents needs 2^k table entries. **Divorcing** inserts an intermediate variable that summarises a group of parents, so the table becomes two small ones instead of one large one.

The intermediate node usually has a meaning of its own — "total load", "combined risk" — which is why the technique improves interpretability as well as size.

> Fewer parameters means fewer numbers a domain expert has to invent.

Speaker notes

Frame parameter reduction as an elicitation problem, not just a storage one: an expert can estimate 4 numbers reliably and 64 numbers not at all.

Modelling Technique: Noisy OR

When many causes $x_1, \ldots, x_k$ can each independently trigger an effect y, a **noisy OR** models the effect as "each cause fires with its own probability, and the effect occurs if any of them fires":

$$p(y = 0 \mid x_1, \ldots, x_k) = \prod_{i:x_i=1} (1 - \lambda_i)$$

k parameters instead of 2^k.

Typical use

Disease–symptom networks, where many diseases give rise to the same symptom.

Speaker notes

Ask what assumption buys the reduction: causes act independently and there is no interaction term. Then ask when that fails — synergistic causes — so the model choice stays a modelling decision.

Uncertain and Unreliable Evidence

Uncertain (soft) evidence

The state of x is itself uncertain, expressed as a distribution $\tilde{x}$. Infer by summing over the states:

$$p(y \mid \tilde{x}) = \sum_{x} p(y \mid x) p(x \mid \tilde{x})$$

We look at the grass without touching it: $p(T{=}1 \mid \tilde{T}) = 0.8$.

Unreliable evidence

The report is kept, but its table is **replaced** by our own judgement of how reliable it is:

$$p(x \mid \mathrm{pa}(x)) \to \tilde{p}(x \mid \mathrm{pa}(x))$$

Jack says his grass is wet, but Jack is in a bad mood and did not check.

Both are drawn with a **dashed** edge.

Speaker notes

The distinction is subtle and worth a moment: uncertain evidence keeps the observation and softens it; unreliable evidence keeps the report but replaces the model of how the report arises with a judgement. Students who conflate them will model rumours as noise.

Takeaway

A belief network is a **DAG plus one conditional table per node**. Modelling techniques — divorcing, noisy OR, soft evidence — all trade assumptions for parameters, and every trade should be stated out loud.

Three Ways Three Nodes Can Connect

Every conditional-independence question in a DAG reduces to these three patterns.

Chain

$$a \to c \to b$$

a and b are dependent — but **independent given** c.

Information flows along the chain until the middle is observed.

Common cause (fork)

$$a \leftarrow c \to b$$

a and b are dependent — but **independent given** c.

The shared cause explains the correlation.

Common effect (collider)

$$a \to c \leftarrow b$$

a and b are **independent** — but **dependent given** c.

Explaining away — the reverse of the other two.

Speaker notes

The whole slide is one table students should be able to reproduce. Emphasise the collider's inversion: conditioning normally destroys dependence, here it creates it. The wet-grass example is exactly a collider at T.

Mini-Exercise

Rain and the sprinkler are independent: rain → wet grass ← sprinkler. Your grass is **wet**, and you learn that it **rained**.

Does that change how likely it is that the sprinkler was on?

Speaker notes

Two minutes, one question, and the one case that surprises. If the group is quick, ask what happens when you condition on a *descendant* of the collider — say, a wet doormat instead of the grass. The same dependence appears, which is why d-separation mentions descendants.

Mini-Exercise

Rain and the sprinkler are independent: rain → wet grass ← sprinkler. Your grass is **wet**, and you learn that it **rained**.

Does that change how likely it is that the sprinkler was on?

Yes — it drops. Rain already explains the wet grass. Observing the common effect made two independent causes dependent: *explaining away*, the collider case. On a chain or a fork, observing the middle does the opposite.

d-Separation

A path between X and Y is **blocked** by a set Z if it contains either

- a chain or fork whose middle node **is in** Z, or
- a collider whose middle node — and all of its descendants — are **not in** Z.

If every path from X to Y is blocked by Z, then X and Y are **d-separated** given Z, and $X \perp\!\!\!\perp Y \mid Z$.

d-separation is sound: what it declares independent really is independent for every distribution factorising over the graph. It is not complete for a *particular* distribution — extra independences may hold numerically without being visible in the graph.

Speaker notes

Definition after the three cases, so it reads as a summary rather than an axiom. The soundness/completeness caveat is the honest version of "graphs have limited expressive power" from the intro unit.

Takeaway

Independence in a belief network is a **graph property**: chains and forks are blocked *by* observing the middle node, colliders are opened by it. d-separation is that rule applied to every path.

The Churn Shop, Drawn

App users churn less (Lecture 02: the gap is real). Two graphs explain it:

FORK

app ← loyalty → churn

Loyal customers install the app *and* stay. The graph claims $\text{app} \perp\!\!\!\perp \text{churn} \mid \text{loyalty}$.

DIRECT EDGE

app → churn ← loyalty

The app itself keeps customers. App and churn stay dependent **given** loyalty.

Speaker notes

The running example (Lecture 01's reasoning-cycle exercise, Lecture 02's test). Both graphs produce the same marginal correlation; they differ in one conditional independence, which is exactly what a missing edge claims. Reading it off: in the fork, loyalty is the middle node, and observing it blocks the path; in the second graph there is a direct edge, which nothing blocks.

Say the limits. Tenure is only a proxy for loyalty, so the check is weaker than the randomised prompt deduced in Lecture 01 — an unmeasured common cause can still be at work. The graph makes the assumption explicit; the data can then contradict it.

The Churn Shop, Drawn

App users churn less (Lecture 02: the gap is real). Two graphs explain it:

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DIRECT EDGE

app → churn ← loyalty

The app itself keeps customers. App and churn stay dependent **given** loyalty.

The graphs disagree on one checkable claim. Compare app users and non-users **of equal loyalty** (say, the same tenure): equal churn favours the fork, a remaining gap the direct edge.

Module 4

Markov Networks

> ***Guiding Question:*** *Pixels in an image influence each other symmetrically — no pixel causes its neighbour. How do we model that?*

Undirected Models

Markov network (Markov random field). An **undirected** graph in which the joint distribution factorises over the **cliques** $c \in \mathcal{C}$ of the graph:

$$p(x_1, \ldots, x_n) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \phi_c(x_c), \qquad Z = \sum_x \prod_c \phi_c(x_c)$$

The $\phi_c \geq 0$ are **potentials** — arbitrary non-negative functions, not probabilities. Z is the normalising **partition function**.

When all cliques have size 2 the model is a **pairwise Markov network**.

Speaker notes

The key difference from a BBN: potentials are not conditional probabilities and do not normalise locally, so a global Z is required. Computing Z is where the cost lives.

Directed vs. Undirected

Belief network

Directed, acyclic. Factors are conditional probabilities. Natural for **causal or generative** stories; conditioning on a collider creates dependence.

Markov network

Undirected. Factors are potentials with a global Z. Natural for **symmetric interactions** — pixels, atoms, neighbours in a lattice.

Speaker notes

Neither subsumes the other: some independence patterns are expressible only in one. Give the standard examples if time permits (a collider is not expressible undirected; a 4-cycle is not expressible directed).

Marginalising and Conditioning Behave Oppositely

Marginalising over C

makes A and B **dependent** — the sum over all states of c couples the potentials.

Conditioning on C

In the chain $A - C - B$, a fixed c is a constant in the potentials, so the factor splits and $A \perp\!\!\!\perp B \mid C$.

This holds because C separates them — not for every C.

Speaker notes

This pair is the undirected counterpart to chains, forks and colliders. In an undirected graph the rule is simple: separation *is* conditional independence, no collider exception.

Markov Properties and the Separation Test

For undirected models, independence is plain graph separation: if every path from X to Y passes through Z, then $X \perp\!\!\!\perp Y \mid Z$.

For a **directed** model the same question is answered by converting the graph:

1. Ancestral graph

Keep $X \cup Y \cup Z$ and their ancestors; delete all other nodes and their edges.

2. Moralisation

Connect any two nodes that share a child, then drop all arrow heads.

3. Separation

Remove the nodes in Z and their edges. If no path joins X to Y, then $X \perp\!\!\!\perp Y \mid Z$.

Speaker notes

"Marrying the parents" is where moralisation gets its name, and it is exactly the collider rule made structural: parents of a common child become connected, so they are no longer separable. This procedure is equivalent to d-separation.

Two Examples

Boltzmann machine

A pairwise Markov network over binary units with weights on the edges; the restricted variant (RBM) allows edges only between a visible and a hidden layer.

Image cleanup

Pixels form a lattice: each potential prefers a pixel to agree with its neighbours and with the noisy observation. Inference denoises the image.

Speaker notes

The image example is the most intuitive Markov network there is — smoothness as a potential. The Boltzmann machine is the bridge to energy-based neural networks, a pointer beyond this material rather than a dependency.

Other Graphical Models

Chain graphs

Combine directed and undirected edges in one model.

Factor graphs

Make each factor an explicit node — mostly used **for inference**, since they state precisely which variables each factor touches.

Undirected models trade causal direction for symmetric interaction, and local normalisation for a global partition function. Separation in the (moralised) graph is the single test for conditional independence.

Module 5

Learning and Inference

> ***Guiding Question:*** *We can now write the model down. Can we actually compute anything with it?*

Three Questions You Can Ask a Model

Infer unobserved variables

Compute the posterior given evidence — probabilistic inference.

Learn the parameters

Estimate the factors' distributions from data.

Learn the structure

Discover which dependencies exist at all.

With complete data, learning the parameters is the easiest of the three — it is counting. Learning the structure is the hardest.

Speaker notes

Structure learning is where causal discovery lives; flag it as an open research area rather than a solved technique, since students will meet confident claims about it.

Inference Is Hard in General

Exact inference in belief networks is **NP-hard**, and so is approximate inference with a guaranteed error bound. Whether a particular model is tractable depends on its structure.

Exact inference

Variable elimination, junction trees, message passing.
Efficient when the graph is a tree or has low treewidth.

Approximate inference

Sampling (MCMC, importance sampling) and variational methods.
Trade the guarantee for tractability.

Speaker notes

The practical rule: structure decides. A chain or tree is linear; a dense graph is hopeless. This is why modelling and inference are not separable concerns — a model you cannot query is not a model.

Learning the Parameters

With **fully observed** data, parameter learning is counting: each conditional table is estimated from the frequencies of its variable and parents.

With **hidden variables**, the counts are unavailable and one alternates between inferring the hidden values and re-estimating the parameters (expectation–maximisation).

> The estimates are only as good as the sample supporting each table row. A parent combination that never occurs in the data leaves its row undetermined — which is another argument for keeping tables small.

Speaker notes

Connect back to divorcing and noisy OR: parameter reduction is also a statistical necessity, not only a storage convenience. The empty row is the same zero-count problem that Laplace smoothing repairs in naive Bayes.

Takeaway

Graphical models make the joint distribution **writable**; whether they make it **queryable** depends on the graph's structure. Parameters come from counting where data is complete, and from iterating where it is not.